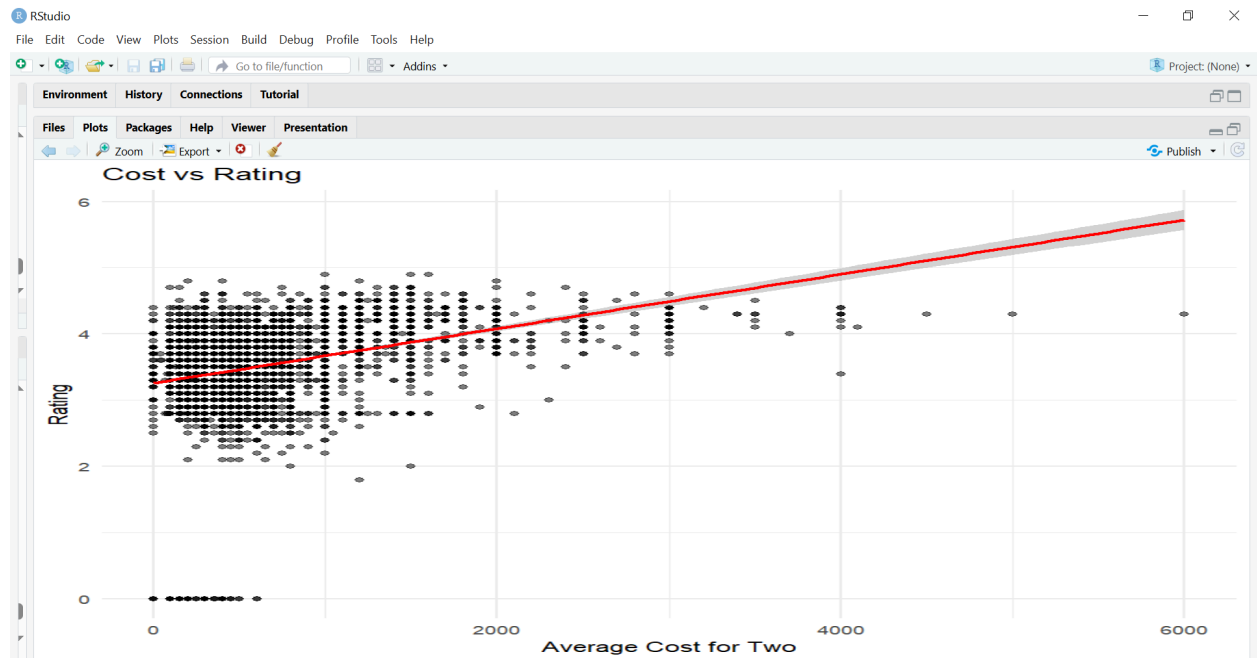
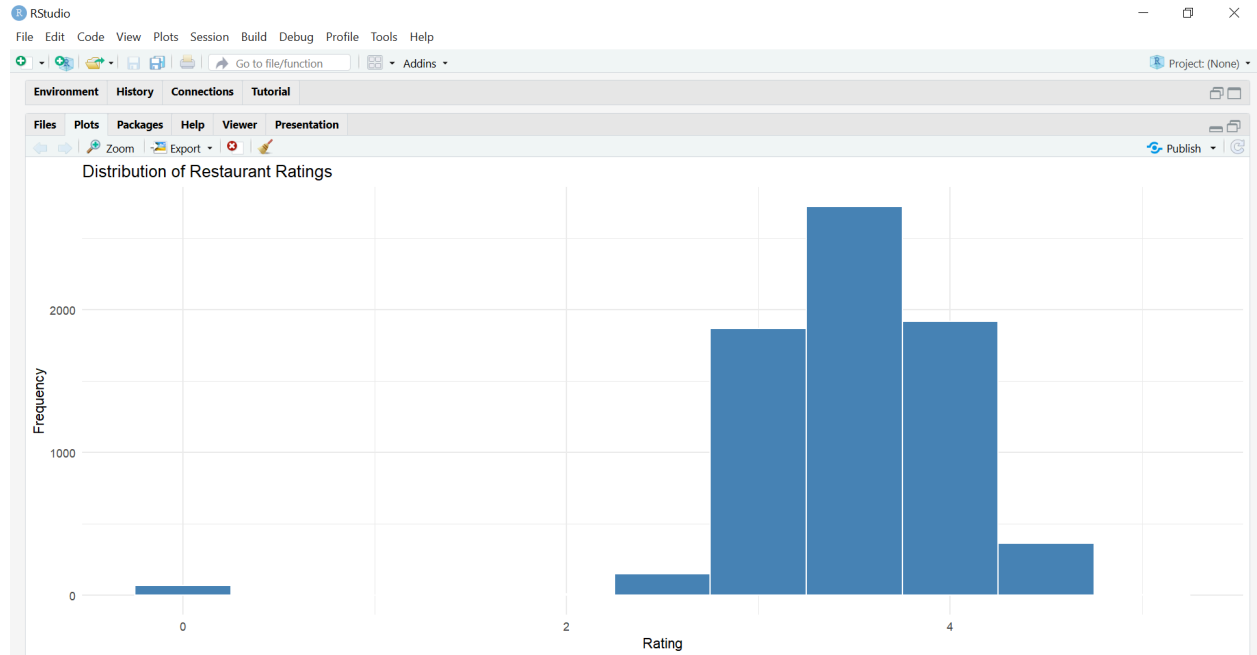
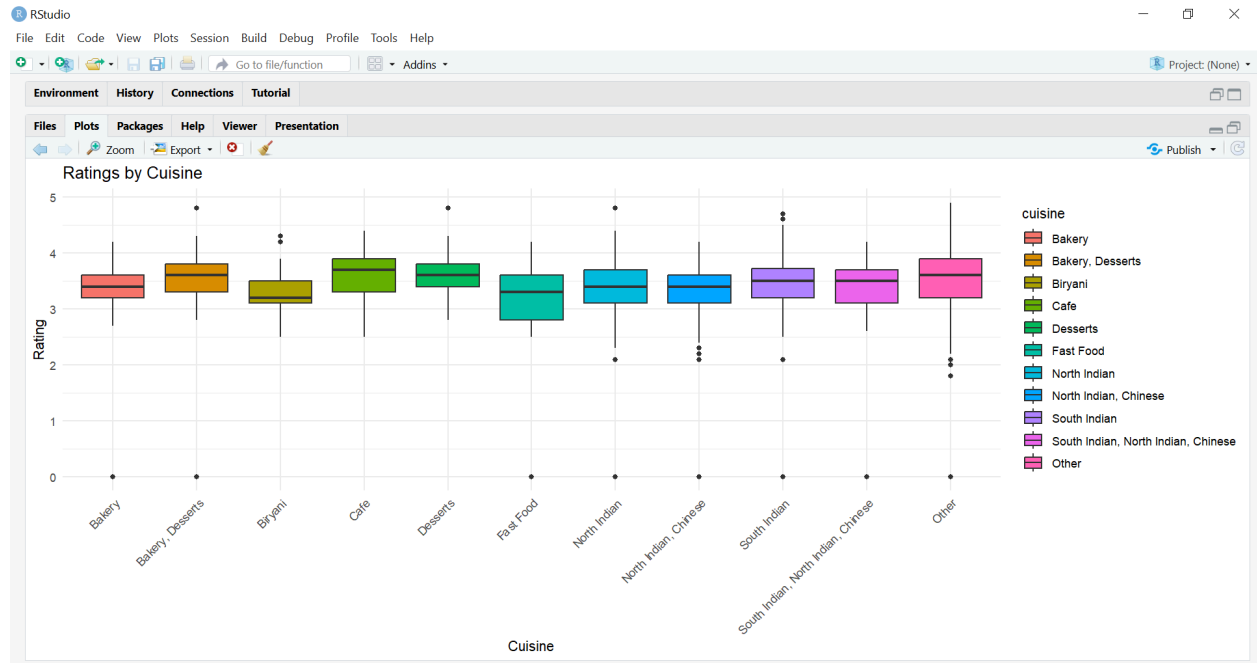


R MODEL



R MODEL



R MODEL

CODE :

```
# =====
# 1. Load Libraries
# =====
packages <- c("tidyverse", "ggplot2", "dplyr", "scales", "janitor", "forcats")
installed <- rownames(installed.packages())
for (p in packages) {
  if (!(p %in% installed)) install.packages(p)
}

library(tidyverse)
library(ggplot2)
library(dplyr)
library(scales)
library(janitor)
library(forcats)

# =====
# 2. Load Dataset
# =====
zomato <- read.csv(
  "Zomato Restaurants Dataset export 2025-09-28 20-08-54.csv",
  stringsAsFactors = FALSE
) %>%
  clean_names() # standardize column names

# =====
# 3. Data Preprocessing
# =====
# Remove unwanted columns safely
zomato <- zomato %>%
  select(-any_of(c("unnamed_0", "unnamed_9")))

# Rename columns consistently
zomato <- zomato %>%
  rename(
    restaurant_name = restaurant_name,
    restaurant_type = restaurant_type,
    rating = rate_out_of_5,
    ratings_count = num_of_ratings,
    avg_cost = avg_cost_two_people,
    cuisine = cuisines_type,
    location = area,
```

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```
    address = local_address
  )

# Helper function for numeric cleaning
clean_numeric <- function(x) as.numeric(gsub(",", "", x))

# Clean numeric columns
zomato <- zomato %>%
  mutate(
    avg_cost = clean_numeric(avg_cost),
    ratings_count = clean_numeric(ratings_count),
    rating = as.numeric(rating)
  )

# Remove missing ratings
zomato <- zomato %>% drop_na(rating)

# Convert categorical columns to factors
zomato <- zomato %>%
  mutate(
    cuisine = as.factor(cuisine),
    location = as.factor(location),
    restaurant_type = as.factor(restaurant_type)
  )

# Lump rare cuisines and locations into "Other"
zomato <- zomato %>%
  mutate(
    cuisine = fct_lump_n(cuisine, n = 10),
    location = fct_lump_n(location, n = 10)
  )

# =====
# 4. Descriptive Statistics
# =====
cat("Summary of Ratings:\n")
print(summary(zomato$rating))

cat("\nTop 10 Cuisine Types:\n")
print(sort(table(zomato$cuisine), decreasing = TRUE)[1:10])

# =====
# 5. Correlation Analysis
# =====
```

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```
cor_cost <- cor(zomato$rating, zomato$avg_cost, use = "complete.obs")
cat("\nCorrelation (Rating vs Cost):", cor_cost, "\n")

cor_reviews <- cor(zomato$rating, zomato$ratings_count, use = "complete.obs")
cat("Correlation (Rating vs Ratings Count):", cor_reviews, "\n")

# =====
# 6. Regression Model
# =====
rating_model <- lm(
  rating ~ avg_cost + ratings_count + cuisine + location,
  data = zomato
)

cat("\nRegression Model Summary:\n")
print(summary(rating_model))

# =====
# 7. Predict Ratings
# =====
zomato <- zomato %>%
  mutate(predicted_rating = predict(rating_model, newdata = zomato))

# =====
# 8. Recommendation System
# =====
zomato <- zomato %>%
  mutate(score = 0.5 * predicted_rating +
    0.3 * rescale(ratings_count) +
    0.2 * (1 - rescale(avg_cost)))

recommendations <- zomato %>%
  arrange(desc(score)) %>%
  select(restaurant_name, cuisine, location, rating, predicted_rating, avg_cost) %>%
  head(10)

cat("\nTop Recommended Restaurants:\n")
print(recommendations)

# =====
# 9. Visualization
# =====
# Histogram: Rating Distribution
ggplot(zomato, aes(x = rating)) +
```

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```
geom_histogram(binwidth = 0.5, fill = "steelblue", color = "white") +  
labs(  
  title = "Distribution of Restaurant Ratings",  
  x = "Rating",  
  y = "Frequency"  
) +  
theme_minimal()
```

Scatter Plot: Cost vs Rating

```
ggplot(zomato, aes(x = avg_cost, y = rating)) +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "lm", color = "red") +  
  labs(  
    title = "Cost vs Rating",  
    x = "Average Cost for Two",  
    y = "Rating"  
  ) +  
  theme_minimal()
```

Boxplot: Ratings by Cuisine

```
ggplot(zomato, aes(x = cuisine, y = rating, fill = cuisine)) +  
  geom_boxplot() +  
  labs(  
    title = "Ratings by Cuisine",  
    x = "Cuisine",  
    y = "Rating"  
  ) +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```